

An overview of systematic reviews on the efficacy and safety of osteopathic techniques

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ABSTRACT

Objective: To investigate the efficacy and safety of manipulative osteopathic techniques in persons of all ages with various medical indications.

Design: Overview of systematic reviews and meta-analyses.

Method: We conducted a systematic literature search for systematic reviews of randomized controlled trials in English or German that compared osteopathic interventions to no treatment, sham treatment or any other non-osteopathic treatment, in bibliographic databases and other sources.

Results: We included 27 systematic reviews covering nine therapeutic indications. Based on moderate quality of evidence, osteopathic treatment was an effective means of reducing pain and partially improving physical function in adults with acute neck pain, chronic non-specific low back pain, low back pain in pregnancy, and chronic non-oncologic pain. It can also shorten the length of hospital stays in premature infants. Furthermore, osteopathic treatment may relieve symptoms in adults with chronic neck pain, postpartum low back pain, migraine, and irritable bowel syndrome, and it may help children with otitis media. It did not appear to provide relief to adults with acute non-specific low back pain, fibromyalgia, headaches of various causes, women with lower urinary tract symptoms, and children with cerebral palsy. Evidence for other indications like vertigo, chronic obstructive pulmonary disease, hypertension, gynecological or birth-related complaints, and a number of pediatric indications was insufficient to draw firm conclusions.

Conclusion: Manipulative osteopathic techniques appears to be particularly effective in musculoskeletal conditions. Overall, the evidence base for the outcomes of manual osteopathic treatment has improved, despite heterogeneous studies that are partly of poor methodological quality.

1. Introduction

Osteopathic medicine is a holistic system of healthcare that integrates patient needs with contemporary practices in medicine, surgery, and obstetrics. It highlights the connection between the body's structure and function, while also acknowledging the body's natural capacity for self-healing (AACOM, 2017).

Diagnosis and therapy are based on five osteopathic models. These represent fundamental categories on which various osteopathic strategies are based (World Health Organization, 2010).

The behavioral model is designed to enhance the biological,

psychological, and social aspects of the health spectrum, including emotional balance and compensatory mechanisms. The metabolic model emphasizes the importance of strengthen the body's self-regulatory and self-healing mechanisms, to promote energy conservation by balancing energy expenditure and exchange, and to enhance immune, endocrine, and organ functions. The neurologic model aims to achieve autonomic balance and manage neural reflex activity. This involves removing facilitated segments, reducing afferent nerve signals, and alleviating pain. In order to influence this aspect of the patient's health, osteopathic manipulative techniques such as counterstrain and Chapman's reflex points are used. The aim of the respiratory-circulatory

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model is to address and improve restrictions in all the diaphragms of the body. These diaphragms are viewed as 'transverse restrictors' that influence motion, venous and lymphatic drainage, and the flow of cerebrospinal fluid. The structural model in osteopathy aims to achieve biomechanical adjustment and joint mobilization by addressing issues in myofascial connective tissues, bony, and soft tissues. Its goal is to eliminate restrictive forces and enhance movement (AACOM, 2017).

Osteopathic medicine encompasses a comprehensive approach to healthcare that incorporates manual techniques as part of its therapeutic arsenal. These hands-on methods, collectively known as Osteopathic Manipulative Treatment (OMT), include a variety of techniques aimed at addressing different aspects of the body's structure and function. Among these are Spinal Manipulative Therapy (SMT), Muscle-Energy Technique (MET), Craniosacral Therapy (CST), Myofascial Release Therapy (MFR), and Visceral Manipulation (VM) (AACOM, 2017). Each technique aims to improve patient outcomes, with varying levels of empirical support; however, their efficacy may differ based on specific applications and contexts.

Spinal manipulative therapy (SMT) treats back and neck pain and other musculoskeletal conditions by applying force to spinal joints. It is a form of manual therapy in which a joint is intentionally moved at high speed, typically towards the limits of its clinical range of motion (Shekelle and Tang, 2023).

Muscle Energy Technique (MET) involves the patient's voluntary muscle contraction in a specific direction against the therapist's counterforce, aiming to lengthen shortened muscles, mobilize restricted joints, strengthen weak muscles, and reduce localized edema and congestion (Franke et al., 2015b).

Craniosacral Therapy (CST) involves gentle, non-invasive palpation techniques applied to the fascia, focusing on the area between the cranium and sacrum. CST focuses on relieving myofascial tension while simultaneously regulating sympathetic nervous system function (Haller et al., 2019).

Myofascial Release Therapy (MFR) is a hands-on technique that uses sustained pressure to mobilize the myofascial complex. MFR aims to change the properties of the connective tissue (myofascial complex) (Ughreja et al., 2021).

Visceral manipulation aims to alleviate pain in the somatic structures that share innervation with the manipulated viscera, reduce adhesions, and enhance the mobility of the viscera. This technique also activates viscerosomatic reflexes and increases abdominal pressure, which may stimulate bowel movements (Switters et al., 2019).

In light of the growing body of systematic reviews on the topic, the aim of our overview is to summarize the results of systematic reviews comparing manual osteopathic techniques alone or as an add-on to other treatments, including further osteopathic approaches, to either no treatment, sham treatment or alternative non-osteopathic treatments, and to assess the efficacy and safety of osteopathic treatments in individuals of all ages and with various medical indications.

2. Methods

This systematic overview has been registered at OSF (registration DOI:10.17605/OSF.IO/VWZGH).

2.1. Inclusion criteria

We included systematic reviews (SRs) and meta-analyses (MAs) that were based on randomized controlled trials (RCTs) because RCTs are the most reliable means of assessing the efficacy of a medical intervention. SRs and MAs that included both RCTs and controlled clinical trials were only included when a sub-analysis for RCTs had been carried out.

Participants could be of any age or gender and have one or several of a wide range of conditions. For example, they could have problems in internal medicine, or orthopedic, pediatric, gynecological, oncologic, neurological, dental and orthodontic, or urological complaints. The

intervention provided to the patients had to involve manipulative osteopathic treatment. Specifically, the following manual treatment methods were considered: osteopathic manipulative treatment (OMT), spinal manipulative therapy (SMT), myofascial release (MFR), visceral manipulation (VOMT or VM), craniosacral therapy (CST) and the muscle energy technique (MET). The interventions could be provided alone or as an add-on to other treatments. The control group included no treatment, sham treatment or any other non-osteopathic treatment of the respective disease/impairment. Outcomes included pain intensity, symptomatology, physical limitations/functionality, quality of life, medication use, and adverse events.

2.2. Search strategy

We conducted a systematic literature search on April 6, 2022, for topically relevant SRs and MAs in bibliographic databases and other sources, without time restrictions. The bibliographic databases were Medline, Embase, CINAHL, Cochrane Database of Systematic Reviews (CDSR) and PEDro. Other sources of information were OSTMED.DR, Osteopathic Research Web, the reference list of the report "Quick Assessment zur Wirksamkeit und Sicherheit osteopathischer Behandlungen für ausgewählte Indikationen" issued by the Austrian National Public Health Institute 2018 (Antony and Eisenmann, 2018), and the publication list of the Austrian Association of Osteopathy. We only included SRs and MAs published in English or German. The database searches were carried out using keywords (MeSH terms and free-text search) associated with osteopathic treatment. Full details of the search strategies are listed in [Supplementary Material A.1](#).

2.3. Study selection

Two review authors independently screened the titles and abstracts of the results identified using the search strategy. The full texts of potentially relevant studies again were independently assessed by two review authors. Disagreements between authors were resolved through discussion or by consulting a third review author.

All publications that fulfilled the defined inclusion criteria were assigned to different indication areas according to the specific research question. If several SRs were available for a specific indication area, we screened and ranked them with regard to their usefulness in the evidence synthesis. Consequently, we prioritized studies for the analysis that best fulfilled the following criteria:

- o Up-to-date
- o Cover various types of osteopathic interventions
- o Include a large number of relevant RCTs
- o Present the results of a meta-analysis of relevant outcomes

2.4. Data extraction and management

Two review authors independently extracted the study data based on a data extraction form. We extracted the following data: author, title, year, journal, study design, aim of the systematic review, sponsoring, inclusion criteria, exclusion criteria, literature search, intervention group, control group, duration of the intervention/follow-up, included relevant RCTs, risk of bias (RoB) of included RCTs, outcomes, results, certainty of evidence, conclusion and limitations. For the evidence synthesis, we used only data obtained directly from SRs, and preferably the results of MAs. If no MAs were available, the results of the included RCTs were summarized descriptively. The relevant outcomes were analyzed and described comparatively and presented separately according to indication area.

2.5. Assessment of the methodological quality of the systematic reviews and quality of evidence

Two authors used the AMSTAR 2 tool to independently assess the methodological quality of the systematic reviews (Shea et al., 2017). We used a consensus method to resolve disagreements. If no agreement could be reached, a third reviewer was consulted.

In addition, the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach (Guyatt et al., 2011) was used to separately rate the certainty of evidence for each condition and outcome.

We used four levels to grade certainty of evidence:

- o High certainty of evidence: the true effect is very likely to be close to the reported effect; further studies are very unlikely to change the result.

- o Moderate certainty of evidence: the true effect is probably close to the reported effect; further studies are unlikely to change the result.
- o Low certainty of evidence: the true effect might be substantially different from the reported effect; further studies are likely to change the result.
- o Very low certainty of evidence: the true effect is likely to be substantially different from the reported effect; further studies are likely to change the result

3. Results

3.1. Results of the literature search

After removing duplicates, the literature search yielded 443 results. We excluded 322 references after screening titles and abstracts. We identified 121 full texts for further examination, of which we considered 76 publications not to be relevant and excluded them (see

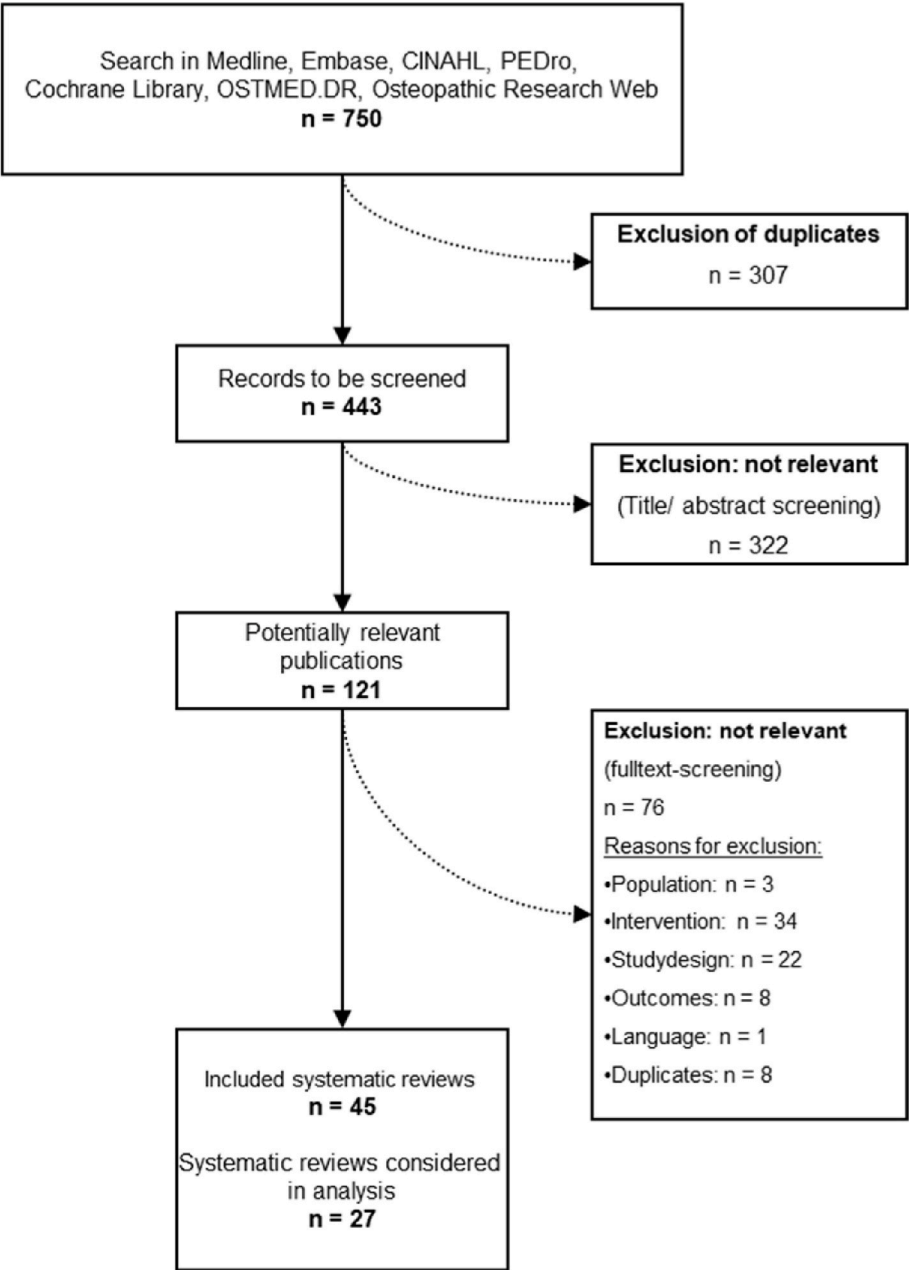


Fig. 1. Study flow diagram.

Supplementary Material A.4). This was carried out as part of the full-text screening process. We considered 45 systematic reviews investigating the efficacy and/or safety of osteopathic treatment in adults or children/adolescents to be relevant, and classified them according to the following therapeutic indications:

- Chronic pain (non-oncologic of any cause)
- Musculoskeletal diseases
- Neurological diseases
- Gastrointestinal diseases
- Urological diseases
- Pulmonary diseases
- Cardiovascular diseases
- Gynecological diseases/impairments
- Pediatric diseases/impairments

No systematic reviews were identified for oncologic or dental and orthodontic conditions. Of the 45 relevant reviews, those that best met the inclusion criteria described in 2.3 were selected for the evidence synthesis. Ultimately, a total of 27 systematic reviews were included in the evidence synthesis (Tramontano et al., 2021; Ughreja et al., 2021; Chen et al., 2021; Rehman et al., 2021; Cote et al., 2021; Dal Farra et al., 2021; Schulze et al., 2020; Rehman et al., 2020; Haller et al., 2019; Baxter et al., 2019; Rist et al., 2019; Parnell Prevost et al., 2019; Franke et al., 2017; Cerritelli et al., 2017; Ruffini et al., 2016; Franke and Hoesle, 2013; Franke et al., 2015b; Muller et al., 2014; Franke et al., 2014; Rubinstein et al., 2012; Proctor et al., 2006; Bronfort et al., 2001; Chaibi et al., 2021; Switters et al., 2019; Raguckas et al., 2016; Franke et al., 2015a; Rubinstein et al., 2011). See Fig. 1 for the study flow diagram. Table S8 in Supplementary Material A.2 shows the number of included systematic reviews and the number of systematic reviews included in the data synthesis and categorized according to the investigated indication areas. The references of the 18 relevant publications that were not included in the evidence synthesis are listed in section A.5 of the Supplementary Material.

3.2. Characterization of the included reviews

The 27 included systematic reviews were based on RCTs and published between 2001 and 2021. Fourteen publications examined OMT (Tramontano et al., 2021; Rehman et al., 2021; Dal Farra et al., 2021; Schulze et al., 2020; Rehman et al., 2020; Parnell Prevost et al., 2019; Franke et al., 2017; Cerritelli et al., 2017; Ruffini et al., 2016; Franke et al., 2014; Muller et al., 2014; Franke and Hoesle, 2013; Raguckas et al., 2016; Franke et al., 2015a), whereby one review (Dal Farra et al., 2021) also examined MFR. Two other reviews included only MFR as an osteopathic technique (Ughreja et al., 2021; Chen et al., 2021). Seven publications dealt with SMT (Cote et al., 2021; Rist et al., 2019; Rubinstein et al., 2011, 2012; Proctor et al., 2006; Bronfort et al., 2001; Chaibi et al., 2021) and two with MET (Baxter et al., 2019; Franke et al., 2015b). The remaining two reviews examined VM (Switters et al., 2019) and CST (Haller et al., 2019).

The majority of systematic reviews only included studies on the osteopathic treatment of adults. One systematic review included studies involving both children/adolescents and adults (Cote et al., 2021), and one systematic review included studies involving only children/adolescents (Parnell Prevost et al., 2019). Most systematic reviews dealt with several osteopathic manual techniques for musculoskeletal diseases or complaints (e.g. non-specific low back pain). Depending on the investigated condition, the number of RCTs included in the SRs varied widely, ranging from two RCTs for vertigo/balance disorders (Tramontano et al., 2021; Proctor et al., 2006) to 26 RCTs for low back pain (Rubinstein et al., 2011). Eighteen of the 27 reviews (67 %) (Tramontano et al., 2021; Ughreja et al., 2021; Chen et al., 2021; Cote et al., 2021; Schulze et al., 2020; Baxter et al., 2019; Rist et al., 2019; Franke et al., 2017; Cerritelli et al., 2017; Ruffini et al., 2016; Muller

et al., 2014; Franke and Hoesle, 2013; Proctor et al., 2006; Bronfort et al., 2001; Chaibi et al., 2021; Switters et al., 2019; Raguckas et al., 2016; Franke et al., 2015a) included fewer than ten RCTs. In 15 of the 27 reviews, the results of the included RCTs were summarized quantitatively based on meta-analyses (Ughreja et al., 2021; Chen et al., 2021; Rehman et al., 2021; Dal Farra et al., 2021; Rehman et al., 2020; Haller et al., 2019; Rist et al., 2019; Franke et al., 2017; Franke et al., 2014; Franke and Hoesle, 2013; Rubinstein et al., 2012; Chaibi et al., 2021; Franke et al., 2015a; Rubinstein et al., 2011; Franke and Hoesle, 2013); the remaining reviews only provided descriptive summaries or comparisons of the results of individual studies. Reasons given for not conducting a meta-analysis were the low number of RCTs, as well as the heterogeneity of the interventions and the heterogeneity of outcome measures (Tramontano et al., 2021; Cote et al., 2021; Schulze et al., 2020; Baxter et al., 2019; Parnell Prevost et al., 2019; Cerritelli et al., 2017; Ruffini et al., 2016; Muller et al., 2014; Proctor et al., 2006; Bronfort et al., 2001; Switters et al., 2019; Raguckas et al., 2016).

Table 1 provides an overview of the included systematic reviews, classified according to indication.

3.3. Results of the quality assessment

Overall, the methodological quality assessment based on the AMSTAR-2 checklist (Shea et al., 2017) showed the 27 included systematic reviews to be of moderate to low methodological quality. The assessment of the methodological quality of each systematic review is presented in Supplementary Material A.3.

3.4. Effect of osteopathic treatment in adults

The results of the systematic reviews and the synthesis of evidence in adults are presented below by indication. Summaries of the effects of the osteopathic interventions compared to control interventions, grouped by indication and osteopathic technique, can be found in Table 2 (Section 3.4.9).

3.4.1. Chronic pain

We analyzed three systematic reviews on chronic non-oncologic pain of any cause published by two groups of authors. The two reviews by Rehman et al. (2020, 2021) focused on the efficacy and safety of OMT as a single therapy compared to sham treatment or standard therapy, or of OMT as an add-on to exercise therapy. Sixteen RCTs involved a total of 1158 patients (Rehman et al., 2020) and 11 RCTs a total of 1015 patients (Rehman et al., 2021). The review by Haller et al. (2019) included ten RCTs and 681 patients receiving CST compared to sham treatment, standard therapy or active controls.

The summarized results of the reviews show that OMT and CST are effective in reducing pain, reducing functional impairment and improving quality of life in patients with chronic non-oncologic pain of various causes, with none or only few side effects. We rated the certainty of evidence for all outcomes as moderate.

3.4.2. Musculoskeletal diseases

Results from 14 systematic reviews were available for musculoskeletal diseases and complaints. Eight reviews reported results for non-specific low back pain (Chen et al., 2021; Dal Farra et al., 2021; Rehman et al., 2020; Franke et al., 2015b; Franke et al., 2014; Rubinstein et al., 2012; Switters et al., 2019; Rubinstein et al., 2011), and three reviews for low back pain in pregnancy (Franke et al., 2014, 2017; Ruffini et al., 2016). Results for fibromyalgia (Ughreja et al., 2021; Schulze et al., 2020) and non-specific neck pain were reported in two systematic reviews (Chaibi et al., 2021; Franke et al., 2015a). No systematic reviews could be identified for other musculoskeletal indications.

Table 1

Characteristics of the included systematic reviews.

Indication area	Author/Publication year	Country	Osteopathic treatment	Number of RCTs
Chronic pain				
Chronic non-oncologic pain of any cause	Rehman et al., /2021 (Rehman et al., 2021)	Canada	OMT	11
	Rehman et al., /2020 (Rehman et al., 2020)	Canada	OMT	16
	Haller et al., /2019 (Haller et al., 2019)	Germany	CST	10
Musculoskeletal diseases				
Non-specific LBP	Dal Farra et al., /2021 (Dal Farra et al., 2021)	Italy	OMT, MFR	10
	Chen et al., /2021 (Chen et al., 2021)	China	MFR	8
	Rehman et al., /2020 (Rehman et al., 2020)	Canada	OMT	7
	Switters et al., /2019 (Switters et al., 2019)	Austria	VM	4
	Franke et al., /2015 (Franke et al., 2015b)	Germany	MET	12
	Franke et al., /2014 (Franke et al., 2014)	Germany	OMT	15
	Rubinstein et al., /2012 (Rubinstein et al., 2012)	USA	SMT	20
	Rubinstein et al., /2011 (Rubinstein et al., 2011)	USA	SMT	26
Pregnancy-related LBP	Franke et al., /2017 (Franke et al., 2017)	Germany	OMT	8
	Ruffini et al., /2016 (Ruffini et al., 2016)	Italy	OMT	4
	Franke et al., /2014 (Franke et al., 2014)	Germany	OMT	5
Fibromyalgia	Ughreja et al., /2021 (Ughreja et al., 2021)	India	MFR	5
	Schulze et al., /2020 (Schulze et al., 2020)	Brazil	OMT	7
Non-specific neck pain	Chaibi et al., /2021 (Chaibi et al., 2021)	Norway	SMT	6
	Franke et al., /2015 (Franke et al., 2015a)	Germany	OMT	3
Neurological diseases				
Headache/migraine	Cote et al., /2021 (Cote et al., 2021)	Canada	SMT	1
	Rist et al., /2019 (Rist et al., 2019)	USA	SMT	6
	Cerritelli et al., /2017 (Cerritelli et al., 2017)	Italy	OMT	5
	Bronfort et al., /2001 (Bronfort et al., 2001)	USA	SMT	9
	Tramontano et al., /2021 (Tramontano et al., 2021)	Italy	OMT	2
Vertigo/balance dis.				
Gastrointestinal diseases				
IBS	Müller et al., /2014 (Müller et al., 2014)	Germany	OMT	5
Urological diseases				
LUTS	Franke et al., /2013 (Franke and Hoesle, 2013)	Germany	OMT	5
Pulmonary diseases				
COPD	Baxter et al., /2019 (Baxter et al., 2019)	Australia	MET	3
	Raguckas et al., /2016 (Raguckas et al., 2016)	USA	OMT	5
Cardiovascular diseases				
Hypertension	Cote et al., /2021 (Cote et al., 2021)	Canada	SMT	2
Gynecological diseases/impairments				
Birth-related complaints	Ruffini et al., /2016 (Ruffini et al., 2016)	Italy	OMT	1
Menopausal complaints	Ruffini et al., /2016 (Ruffini et al., 2016)	Italy	OMT	1
Dysmenorrhoea	Cote et al., /2021 (Cote et al., 2021)	Canada	SMT	1
	Ruffini et al., /2016 (Ruffini et al., 2016)	Italy	OMT	2
	Proctor et al., /2006 (Proctor et al., 2006)	NZ	SMT	2
Pediatric diseases/impairments				
Infantile colic	Cote et al., /2021 (Cote et al., 2021)	Canada	SMT	1
	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1
Pediatric voiding dysf.	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1
Temporomandibular dis.	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1
Asthma	Cote et al., /2021 (Cote et al., 2021)	Canada	SMT	1
	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1
Obstructive apnoea	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1
Otitis media	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	3
ADHD	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1
Cerebral palsy	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	3
Premature infants	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	3
Postural asymmetry	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1
Scoliosis	Parnell Prevost et al., /2019 (Parnell Prevost et al., 2019)	USA	OMT	1

ADHD: Attention-deficit/hyperactivity disorder; COPD: chronic obstructive pulmonary disease; CST: Craniosacral therapy; dis.: disorders; dysf.: dysfunctional; IBS: Irritable bowel syndrome, LUTS: lower urinary tract symptoms; LBP: low back pain; MET: Muscle-Energy Technique; MFR: Myofascial release therapy; NZ: New Zealand; OMT: Osteopathic manipulative treatment; RCT: Randomized controlled trial; SMT: Spinal manipulative therapy; USA: United States of America, VM: Visceral manipulation.

3.4.2.1. Non-specific low back pain. The efficacy and safety of osteopathic treatments (OMT or MFR alone) were investigated in the Dal Farra et al. (2021) review, which included ten RCTs and 1160 participants. Results on OMT were found in Franke et al. (2014) with 15 RCTs and 1502 patients, and Rehman et al. (2020) with seven RCTs and 742 participants. Another review by Franke et al. (2015b) examined MET and included a total of 12 RCTs with 500 participants. SMT was

investigated in the two Cochrane reviews by Rubinstein, which included 20 RCTs and 2674 participants on acute low back pain (Rubinstein et al., 2012) and 26 RCTs with 6070 participants on chronic low back pain (Rubinstein et al., 2011). Chen et al. (2021) included eight RCTs and a total of 386 patients receiving MFR. Switters et al. (2019) included four RCTs and 303 participants receiving VM. The control groups involved no treatment, sham therapy and/or active controls.

Overall, the available evidence shows that OMT, MFR and SMT reduced pain in patients with chronic non-specific low back pain. We found no statistically significant effect on pain intensity for MET alone. The results on a reduction in functional impairment are inconsistent. We considered the overall certainty of the evidence to be moderate for all outcomes.

For non-specific acute low back pain, the available evidence does not indicate that either SMT or MET were efficacious, and the overall certainty of the evidence was rated as low.

There is no increased potential for harm in patients with non-specific low back pain from osteopathic treatment compared to no treatment or other forms of therapy.

3.4.2.2. Pregnancy-related low back pain. Franke et al. (2017) investigated the effectiveness and safety of OMT for pregnancy-related low back pain in a review involving a total of eight RCTs and 862 participants. Results on OMT were also found in the reviews by Franke et al. (2014) with five RCTs and 361 participants and Ruffini et al. (2016) with four RCTs and 515 participants.

Overall, the available evidence from the reviews show that OMT is effective in reducing pain and functional impairment in women with low back pain in pregnancy or postpartum. We rated the certainty of evidence for low back pain in pregnancy as moderate for all outcomes but as low for post-partum low back pain.

Information about adverse events was only available from Franke et al. (2017), who reported that no adverse events occurred in the included RCTs. There is no increased potential for harm from osteopathic treatment compared to no treatment or other forms of therapy for women with pregnancy-related low back pain.

3.4.2.3. Non-specific neck pain. For neck pain, we identified one systematic review for OMT, which included a total of three RCTs and 123 participants (Franke et al., 2015a), and one for SMT, which included six RCTs and 446 participants (Chaibi et al., 2021).

Overall, the available evidence shows that OMT and SMT are effective in reducing pain in patients with non-specific acute or chronic neck pain compared to the control groups. The certainty of the evidence was rated as low for OMT and moderate for SMT. There was no evidence of the efficacy of OMT or SMT in influencing functional impairment and quality of life. There were no serious adverse events in the RCTs on OMT and SMT. Mild adverse events were rare, with no differences between study groups. Osteopathic treatment does not increase the potential for harm in patients with non-specific neck pain, compared to no treatment or other forms of therapy.

3.4.2.4. Fibromyalgia. Two systematic reviews were available for fibromyalgia, both of which examined the effectiveness of MFR compared with no treatment, sham therapies or active interventions. Schulze et al. (2020) reported results from a total of seven RCTs and involved 368 participants, Ughreja et al. (2021) included five RCTs and 267 participants.

Based on the available evidence, MFR is not effective in reducing pain, or the impact of fibromyalgia, on the daily lives and quality of life of patients with fibromyalgia. Results on adverse events were not reported in the two reviews. The overall certainty of evidence was rated as low for all outcomes.

3.4.3. Neurological diseases

Five systematic reviews investigated the effect of osteopathic treatment on headache, migraine or vertigo/balance disorders (Tramontano et al., 2021; Rist et al., 2019; Cerritelli et al., 2017; Bronfort et al., 2001;

Cote et al., 2021). For other neurological indications, no systematic reviews could be identified.

3.4.3.1. Headache or migraine. Cerritelli et al. (2017) included five RCTs on OMT compared to standard treatment, sham therapies or active interventions and involved a total of 265 participants. Two other SRs investigated the efficacy of SMT compared to sham therapy or other treatments and included nine RCTs and 683 participants (Bronfort et al., 2001) and six RCTs and 677 participants (Rist et al., 2019) respectively. The review by Cote et al. (2021) included an RCT on migraine, but the RCT was already included in the Rist et al. (2019) meta-analyses and therefore provided no additional data.

The evidence shows that OMT and SMT are not effective in reducing pain or medication intake in patients with headaches of various causes. However, the available evidence for migraine indicates that SMT is effective in relieving pain from migraine. The overall certainty of evidence was assessed as low for all outcomes. The available evidence showed no increased potential for harm from osteopathic treatment compared to no treatment or other forms of therapy for patients with headache.

3.4.3.2. Vertigo. The systematic review by Tramontano et al. (2021) addressed OMT and included two RCTs and involved a total of 40 participants with vertigo/balance disorders. The positive effects of OMT compared to no treatment or sham treatment were statistically significant for individual outcomes, but the overall efficacy of OMT was unclear, partly due to the very limited available evidence. No information was provided on adverse events. Based on the available evidence, the results showed that OMT does not reduce impairment to daily living in patients with vertigo/balance disorders. The evidence was insufficient to permit a reliable assessment of the efficacy of osteopathic treatment in patients with vertigo.

3.4.4. Gastrointestinal diseases

The systematic review by Muller et al. (2014) compared OMT to sham treatment in five RCTs and a total of 204 participants with irritable bowel syndrome (IBS). No systematic reviews could be identified for any other gastrointestinal indications. The available evidence showed that OMT is potentially effective in relieving pain in patients with IBS. The overall certainty of the evidence was rated as low.

3.4.5. Urological disorders

Franke and Hoesle (2013) investigated OMT in women with lower urinary tract symptoms (LUTS) such as urinary incontinence or bladder voiding dysfunction and included five studies and a total of 233 participants. No systematic reviews could be identified for any other urological indications.

Based on the available evidence, OMT is not effective in reducing symptoms in women with lower urinary tract disorders. The certainty of the evidence was assessed as low. Results from studies with men with symptoms of the lower urinary tract were not available.

3.4.6. Pulmonary diseases

Two systematic reviews investigated the effect of osteopathic treatments on COPD. Raguckas et al. (2016) examined five RCTs comparing OMT's ability to improve lung function with standard treatments, sham treatments and active interventions in 157 participants. Baxter et al. (2019) investigated the efficacy of MET in COPD and included three RCTs and 90 participants. No systematic reviews could be identified for any other pneumological indications.

Overall, the results show that OMT and MET does not improve lung

function or exercise tolerance in patients with COPD. The overall certainty of the evidence was considered to be low to very low.

According to available evidence, osteopathic treatments do not increase the potential for harm in patients with COPD, compared to no treatment or other forms of therapy.

3.4.7. Cardiovascular diseases

Cote et al. (2021) reported results on SMT based on two RCTs and a total of 101 hypertensive participants. No systematic reviews could be identified for any other cardiovascular indications.

Overall, SMT did not influence blood pressure or improve general health or stress levels in hypertensive participants. The evidence was insufficient to permit a reliable assessment of the efficacy of osteopathic treatment in this patient group.

3.4.8. Gynecological diseases/impairment

Three systematic reviews investigated the effect of osteopathic treatments on gynecological diseases/impairments. Ruffini et al. (2016) included eight RCTs on OMT for different indications, including four RCTs on low back pain in pregnancy. Four other RCTs, which involved a total of 155 participants, investigated the effectiveness of OMT in comparison to standard therapy or sham treatment for childbirth-related problems, menopausal symptoms and dysmenorrhea. Dysmenorrhea was also investigated in another review by Proctor et al. (2006). The review included two RCTs on SMT and a total of 183 patients. Cote et al. (2021) also reported results on SMT from an RCT with 69 dysmenorrhea patients. No systematic reviews could be identified for further gynecological indications. Meta-analyses were not performed for any of the three reviews.

Overall, despite significant results in individual cases, OMT and SMT had no clear influence in patients with dysmenorrhea or women with childbirth-related problems and menopausal symptoms. The evidence was insufficient to permit a reliable assessment of the efficacy of osteopathic treatment in such cases.

SMT does not appear to increase the potential for harm, compared to sham treatments for patients with dysmenorrhea.

3.4.9. Summary of the results in adults

Table 2 provides an overview of the overall effects and certainty of evidence for the results of osteopathic treatment in adults.

3.5. Effect of osteopathic treatment in children

The systematic review by Parnell Prevost et al. (2019) focused on OMT for various pediatric conditions and included 18 RCTs and a total of 1704 children/adolescents. Another review, Cote et al. (2021), reported the results of one RCT investigating the effect of SMT on infantile colic and one RCT on infantile asthma. Overall, results were provided for 11 different indications, although only one RCT with a very low number of participants was available for many indications.

Overall, the available evidence shows that compared to no treatment, OMT reduces the length of hospital stays in premature infants and reduces the frequency of illness in children with otitis media. The certainty of the evidence was rated as moderate or low. Furthermore, OMT did not improve the health of children with cerebral palsy, whereby the certainty of the evidence was low. For other investigated pediatric indications, too little evidence exists to assess the efficacy of osteopathic treatments (OMT, SMT).

Table 3 provides an overview of the overall effects and certainty of evidence for the results of osteopathic treatment for pediatric indications.

4. Discussion

4.1. Summary of the main results

In our analysis, we included 27 systematic reviews comparing the efficacy and safety of various osteopathic manipulative treatments with control interventions in persons of all ages. In addition to the 27 systematic reviews included in the evidence synthesis, a further 18 relevant systematic reviews were identified during the literature search but were not included in the analyses. The main reasons for exclusion were a publication date in the distant past or a very general research question that would have cast doubt on whether the approach taken in the literature search for osteopathic interventions could be considered systematic. However, a comparison of the content of these reviews did not reveal any significant differences compared with the results of the papers included in the analyses conducted in our systematic overview.

Our overview of systematic reviews based on RCTs revealed moderate certainty of evidence for the positive effects of osteopathic manipulative treatment in improving pain in adults with acute neck pain, chronic non-specific low back pain, low back pain in pregnancy, and chronic non-oncologic pain. Similarly, osteopathic treatment can improve physical functionality in individuals with chronic non-oncologic pain and low back pain in pregnancy, based on moderate certainty of evidence. OMT can also shorten the length of hospital stays in premature infants.

The certainty of evidence that osteopathic treatment may improve symptoms in adults with chronic neck pain, postpartum low back pain, migraine, and irritable bowel syndrome, as well as in children with otitis media was low, as was evidence that physical functionality may improve in women with postpartum low back pain.

In adults with acute non-specific low back pain, fibromyalgia, headaches of various causes, LUTS or in children with cerebral palsy, osteopathic treatment appeared to have no effect on symptoms. However, the certainty of the evidence was low.

For other indications like vertigo, COPD, hypertension, gynecological or birth-related complaints, or for several pediatric indications, current evidence on the effectiveness of osteopathic treatment was insufficient. The systematic reviews provided no information on the effectiveness of osteopathic treatment of oncologic, as well as dental and orthodontic conditions.

Overall, none of the included systematic reviews reported a significant increase in the rate of adverse events following osteopathic manipulative treatment, as compared with control interventions. Serious adverse events occurred very rarely in all the included systematic reviews.

Implication of research arise from the still insufficient or missing evidence on the effectiveness of osteopathic treatment for some of the indications of interest. Most of the studies, especially which had insufficient evidence on the effectiveness of osteopathic treatment, included fewer than 200 patients. To obtain more robust results, more RCTs with a larger sample size would be needed. In addition, the interventions in the studies are neither consistent nor clearly structured, which limits the comparability of the interventions. Therefore, there is a need for studies with longer periods of follow-up and with comprehensible and well-described interventions. This could also help to identify potential variations in effectiveness among different groups of practitioners.

4.2. Methodological quality

Overall, the methodological quality of the included reviews was rated as moderate to low based on the AMSTAR 2 checklist. The main reasons for assessments of low quality were a limited search strategy, a

Table 2

Chronic pain, musculoskeletal and other indications in adults: Overview of overall effects and certainty of evidence.

Indication Osteopathic treatment		Effect (certainty of evidence)				AEs
		Pain intensity	Functional status	Quality of life	Other efficacy outcomes	
Chronic non-oncologic pain of any cause	OMT	↑ (mod)	↑ (mod)	↑ (mod)	–	⇔ (mod)
	CST	↑ (mod)	↑ (mod)	↑ (mod)	–	⇔ (mod)
Chronic nonsp. LBP	OMT	↑ (mod)	↑↓ (mod)	–	–	–
	SMT	↑ (mod)	↑ (mod)	↑↓ (low)	–	–
	MFR	↑ (mod)	↑↓ (v.low)	–	–	–
	MET	⇔ (low)	⇔ (low)	–	–	–
Acute nonsp. LBP	SMT	↑↓ (low)	↑↓ (low)	–	–	⇔ (low)
	MET	⇔ (low)	⇔ (low)	–	–	–
LBP in pregnancy	OMT	↑ (mod)	↑ (mod)	–	–	⇔ (mod)
Post-partum LBP	OMT	↑ (low)	↑ (low)	–	–	–
Chronic nonsp. neck pain	OMT	↑ (low)	⇔ (low)	–	–	⇔ (low)
Acute nonsp. neck pain	SMT	↑ (mod)	↑↓ (low)	⇔ (low)	–	⇔ (mod)
Fibromyalgia	OMT	○	○	–	–	–
	MFR	↑↓ (low)	–	↑↓ (low)	fibromyalg. impact: ↑↓ (low)	–
	OMT	↑↓ (low)	–	–	medication intake: ↑↓ (low)	⇔ (low)
Headaches of various causes	SMT	↑↓ (low)	–	–	medication intake: ↑↓ (low)	–
	OMT	↑ (low)	–	–	medication intake: ↑ (low)	⇔ (low)
Migraine	SMT	↑ (low)	–	–	medication intake: ⇔ (low)	⇔ (low)
	OMT	–	–	–	impairment: ↑↓ (v.low)	–
Vertigo/balance disorder	OMT	↑ (low)	–	–	IB severity: ↑↓ (v.low)	–
IB Syndrome	OMT	–	–	–	symptoms: ↑↓ (low)	–
LUTS	OMT	–	–	–	lung function: ↑↓ (low)	⇔ (low)
COPD	OMT	–	–	–	physical strength: ↑ (low)	–
					lung function: ⇔ (v.low),	⇔ (v.low)
					physical strength: ↑ (v.low)	
					symptoms: ↑ (v.low),	
Hypertension					shortness of breath: ↑ (v.low)	
	SMT	–	–	–	blood pressure: ⇔ (v.low)	○
					stress: ⇔ (v.low)	
Dysmenorrhea	OMT	↑ (v.low)	–	–	general health: ⇔ (v.low)	–
	SMT	↑↓ (v.low)	–	–		⇔ (v.low)
Birth-related complaints	OMT	–	–	–	episiotomy: ↑ (v.low)	–
					PT and CS: ⇔ (v.low)	
Menopausal complaints	OMT	–	–	–	hot flushes: ↑ (v.low)	–
					psychosomatic: ⇔ (v.low)	

↑: Positive effect of osteopathic intervention compared to control; ⇔: no effect of osteopathic intervention compared to control, homogeneous results; ↑↓: no effect of osteopathic intervention compared to control, heterogeneous result; ○: insufficient data; –: outcome not reported.

AEs: adverse events; CS: Cesarean Section; CST: Craniosacral Therapy; disord: disorders; fibromyalg.: fibromyalgia; IB: Irritable bowel; LBP: low back pain; LUTS: Lower urinary tract symptoms; MET: Muscle Energy-Technique; MFR: Myofascial-release therapy; mod: moderate; nonsp.: nonspecific; OMT: Osteopathic Manipulative Treatment; PT: Perineal tear; SMT: Spinal-manipulative therapy; v.low: very low.

limited description of the characteristics of the included RCTs and a failure to conduct meta-analyses without adequate justification. Only slightly more than half of the included systematic reviews included results from meta-analyses. The risk of bias (RoB) of the RCTs included in the reviews was mostly high to moderate. The main reasons for high RoB were inadequate descriptions of the randomization process, concealment of group allocation, and a lack of application of the intention-to-treat (ITT) principle in the analysis. Blinding of treatment providers was generally absent, and blinding of patients was only partial because the nature of the interventions largely precluded blinding. Only 26 % of the included RCTs showed a low RoB. Overall, there was no significant difference in the RoB of the RCTs in the various indication areas, and only neck pain and pediatric indications had a comparatively high proportion of RCTs with low RoB. To improve the quality of future SRs and RCTs, we recommend that authors adhere to existing guidelines and recommendations, such as the Cochrane Handbook (Higgins and Thomas, 2024) and the CONSORT statement (Schulz et al., 2010). We also emphasize the importance of implementing appropriate blinding procedures, or alternatively, using objective outcome measures, blinded

outcome assessment, or sensitivity analyses to minimize bias, particularly in cases where complete blinding is not feasible.

4.3. Strengths and limitations

This overview of systematic reviews provides a comprehensive and authoritative overview of the effectiveness and risks of osteopathic treatment across the lifespan, identifying any age-specific trends or gaps in the current evidence and informing future research and practice.

The data employed in our analyses was derived exclusively from systematic reviews. Therefore, we are inherently dependent on the criteria defined by the authors of these reviews, including their definitions of the intervention and the control intervention for the RCTs. These criteria vary across the included reviews. Some systematic reviews did not allow any additional interventions in the intervention groups. Others permitted additional interventions without further specification (Rehman et al., 2020; Chaibi et al., 2021), while some allowed only predefined additional interventions (Chen et al., 2021; Ughreja et al., 2021) in the intervention group. The same is true for the control groups,

Table 3
Pediatric indications: Overview of overall effects and certainty of evidence.

Indication Osteopathic treatment		Effect (certainty of evidence)	
		Efficacy outcome	AEs
Infantile colic	OMT	crying time: ○	○
	SMT	crying time: ⇌ (v.low)	–
Pediatric dysfunctional voiding	OMT	symptoms: ↑ (v.low)	–
Temporomandibular dysfunction	OMT	function: ↑ (v.low)	–
Asthma	OMT	lung function: ○	–
	SMT	lung function: ⇌ (v.low)	–
Obstructive apnoea	OMT	frequency: ○	–
Otitis media	OMT	frequency: ↑ (low)	⇌ (v. low)
ADHD	OMT	accuracy and rapidity: ↑ (v. low)	–
Cerebral palsy	OMT	function: ↑↓ (low)	○
		quality of life: ↑↓ (v.low)	
Premature infants	OMT	length of hospital stay: ↑ (moderate)	○
Postural asymmetry	OMT	asymmetry: ⇌ (v.low)	⇌ (v. low)
Scoliosis	OMT	flexibility: ⇌ (v.low)	⇌ (v. low)

↑: Positive effect of osteopathic intervention compared to control.
⇌: no effect of osteopathic intervention compared to control, homogeneous results.
↑↓: no effect of osteopathic intervention compared to control, heterogeneous results.
○: insufficient data-: outcome not reported.
ADHD: Attention deficit hyperactivity disorder; AEs: Adverse events; OMT: Osteopathic Manipulative Treatment; SMT: Spinal-manipulative therapy; v.low: very low.

with some reviews allowing a wide range of control interventions, such as including sham, standard care, physical therapy, and no treatment (Franke et al., 2014) or spinal mobilization, manual mobilization techniques, medication, and physiotherapy (Chaibi et al., 2021).

The included reviews do not provide a clear distinction between Osteopathic Manipulative Treatment (OMT) in its strictest sense as practiced by osteopaths and manipulative osteopathic techniques performed by other professionals. As a result, the findings cannot be specifically attributed to any particular group of practitioners, leaving it unclear whether there are practitioner-specific differences.

The largest number of systematic reviews based on RCTs concern osteopathic treatment of musculoskeletal diseases, whereby non-specific low back pain was investigated most frequently in the systematic reviews we found. In addition to musculoskeletal diseases, osteopathic treatment of other conditions was also covered in this overview of systematic reviews. The other investigated indications were chronic pain, pediatric disorders, neurological, gastrointestinal, urological, pulmonary, cardiovascular and gynecological diseases. Our overview also provides results on osteopathic treatment in adults as well as in children. Despite the advantage of comprehensiveness, one limitation is that few RCTs were included for the other indications. We also investigated various osteopathic techniques such as MRT, MET, VM, CST or SMT. Although the investigated interventions were comprehensive, a further limitation is that they were heterogeneous in terms of the employed techniques, as well as the frequency and duration of the interventions in the individual RCTs. The RCTs investigated osteopathic treatments both as stand-alone therapies and add-ons to standard treatments, or other

mostly non-drug treatments such as exercise programs or physiotherapy. While such variability of interventions is inherent in manual therapies and osteopathic medicine in particular, it makes it difficult to compare the various studies, which, in turn, makes it difficult to generalize the results. Furthermore, the descriptions of the interventions in the reviews were sometimes very vague. Control interventions also varied widely between the RCTs. In some RCTs, treatments like physiotherapy, massage or exercise programs served as control interventions, whereas in others, controls were sham interventions or no treatment.

Additionally, the limited number of RCTs for a wide array of medical indications restricts current research on OMT. This limitation hinders the ability to conduct high-quality meta-analyses that could accurately determine the efficacy of OMT on its own or relative to specific control interventions. To strengthen future research, we recommend conducting more RCTs that use sham treatment as controls and evaluate the effects of osteopathic manipulative techniques in isolation, without combining them with other interventions. This would not only clarify the attribution of observed effects to OMT, but also contribute to a more consistent evidence base, allowing for more robust meta-analyses across a wider range of indications.

Our analysis is only based on results from systematic reviews. There may be conditions for which the efficacy of OMT has only been investigated in very few RCTs and which were therefore not included in our publication.

4.4. Comparison with a similar overview of systematic reviews: agreement and disagreement

One recently published overview of systematic reviews by Bagagiolo et al. (2022) reported on the efficacy and safety of OMT for various conditions. It included SRs and MAs based on RCTs and included participants of any age and gender that had received OMT. All clinically relevant outcomes were considered, and the review included six systematic reviews addressing musculoskeletal conditions and one each on neurological, pediatric and visceral conditions. Of the nine SRs, seven SRs were also included in our systematic overview, while we included 20 additional SRs and also took MFR, MET, VOMT or VM, CST and SMT into account as further osteopathic treatments worthy of consideration.

Bagagiolo et al. (2022) showed that OMT was more effective than control interventions in reducing pain and improving functional status in patients with acute/chronic non-specific low back pain, chronic non-specific neck pain and chronic non-cancer patients. However, the evidence of OMT efficacy in the treatment of pediatric conditions, irritable bowel syndrome and primary headache was rated as weak because of small sample sizes, conflicting results and/or heterogeneity.

Similar to Bagagiolo et al. (2022), we also found osteopathic treatments (OMT, MFR, SMT) to have a positive effect in case of chronic pain (non-specific low back or neck pain and non-oncologic pain), compared to the control group on pain improvement. However, in contrast to that study, our study showed osteopathic treatments (SMT or MET) to have no effect on pain or functionality in patients with acute non-specific low back pain. While in Bagagiolo et al. (2022), studies that assessed the effect of OMT on persons with acute and chronic non-specific low back pain were combined, we focused on studies investigating the effect of osteopathic treatment (SMT, MET) on individuals whose sole complaint was acute non-specific low back pain, which might explain the differing results.

Both our study and Bagagiolo et al. (2022) identified an improvement in functional status in patients with chronic non-specific low back pain based on studies on OMT and MFR that included individuals with

acute/chronic non-specific low back pain or low back pain in pregnancy. In our study, the effect of osteopathic treatment in persons with particularly chronic non-specific low back pain was inconsistent in terms of functional status. In Bagagiolo et al. (2022) the assessment of persons with acute as well as chronic non-specific low back pain or non-specific low back pain in pregnancy was combined, while we included a subgroup analysis in our study. This and the inclusion of a wider selection of osteopathic manual techniques in our study resulted in some discrepancies between the two overviews.

We agreed with Bagagiolo et al. (2022) that the effects of osteopathic treatments on headache were heterogeneous. Both overviews analyzed Cerritelli et al. (2017) on OMT, the results of which varied considerably. We included a second SR with a larger number of subjects ($n = 683$) on SMT but the results of this review were also heterogeneous. We also separately analyzed the effect of osteopathic treatments on migraine, whereby the available evidence suggests that SMT is effective in reducing the intensity of migraine attacks.

In addition to our overview, Bagagiolo et al. (2022) included the SR of pediatric conditions by Posadzki et al. (2013). This SR showed no conclusive evidence on the efficacy of OMT for any pediatric conditions due to the low methodological quality of included RCTs and other limitations. In our study, the evidence shows OMT to be more effective than no treatment in reducing the length of hospital stays of premature infants (moderate certainty of evidence) and in reducing morbidity in children with otitis media (low certainty of evidence), based on the more recent SR of Parnell Prevost et al. (2019).

In contrast to Bagagiolo et al. (2022), our study examined further indications such as vertigo, LUTS, COPD, hypertension and gynecological conditions. In both overviews, no adverse events were reported in most SRs.

5. Conclusion

Most existing systematic reviews of osteopathic treatments focus on musculoskeletal conditions and especially on non-specific low back pain. Osteopathic techniques (OMT, MFR, SMT) can reduce pain in adults with musculoskeletal conditions, particularly chronic non-specific low back pain or pregnancy-related low back pain. In addition, osteopathic treatment can be effective in reducing acute neck pain and may be effective in reducing chronic neck pain. Osteopathic treatment does not appear to be effective in reducing acute non-specific low back pain. Osteopathic manipulative treatments do not appear to increase the potential for harm.

Only a few systematic reviews of osteopathic treatment for other indications are available, and they tend to involve a low number of RCTs and a limited number of participants. Nonetheless, osteopathic treatment may relieve pain caused by migraine and irritable bowel syndrome. It may also lead to a reduction in the length of hospital stays in premature infants, as well as a reduction in the frequency of illness in children with otitis media. On the other hand, it may not relieve symptoms of headache of various causes, fibromyalgia, LUTS, in adults, and cerebral palsy in children. In case of vertigo, COPD, hypertension, gynecological or birth-related complaints, and various pediatric conditions, insufficient evidence is available to perform a reliable assessment of the efficacy of osteopathic treatment, and further research based on high-quality studies will be required before reliable conclusions can be drawn.

Overall, despite heterogeneity in the quality of studies, reviews and RCTs, the amount of evidence on the efficacy of osteopathic treatment has improved considerably in recent year, with potential for further insights by differentiating the professional background of those administering these techniques.

CRedit authorship contribution statement

Carolyn R. Zipp: Writing – original draft, Data curation. **Thomas Semlitsch:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization. **Gregor Tögel:** Methodology, Data curation. **Cornelia Krenn:** Writing – review & editing, Data curation. **Christine Loder:** Writing – review & editing, Data curation. **Klaus Jeitler:** Writing – review & editing, Methodology. **Andrea Siebenhofer:** Writing – review & editing, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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We have used Perplexity or ChatGPT to check grammar and spelling in some places. All proposed changes were manually reviewed and adjusted as necessary by the authors.

Appendix A. Supplementary data

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